

All polynomials are functions but not all functions are polynomials.

If a function is a polynomial, it is always continuous.

$$ex \circ f(x) = -x^3 + 3x^2 - 2x + 5$$

Continuity conditions:

- 1) Limit must exist
- 2) Value must exist
- 3) Limit value = Exact value

Rolle's theorem:

- 1) $f(x) \in$ continuous $[a, b]$ closed interval
- 2) $f'(x)$ exists in (a, b) open interval
- 3) $f(a) = f(b)$

Then $f'(c) = 0$, $c \in [a, b]$

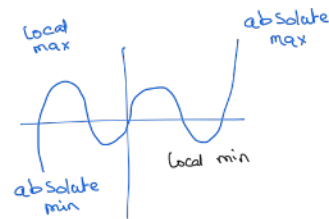
there could be more than 1 c

Extreme value theorem:

Where $f'(x) = 0$, it is called a critical point which can be max or min

local max / local min can be in multiple points

absolute max / absolute min are just one point, either the highest or lowest point



Steps to find critical point:

- 1) Find the derivative
- 2) make the derivative equal to 0
- 3) a) if the value we found is within the interval, accept it
b) if the value isn't within the interval, reject it
- 4) calculate f value that the critical points add end points or the interval

Intermediate value theorem:

If the function is continuous and is going from positive to negative on the y-axis, there will be at least one point where the x-axis is intersected or 0 is at x



Taylor's theorem:

Converts any function to a polynomial approximately

The degree will be given to know when you should stop the formula

You can use polynomials as functions but the degree of x won't change

$$P_n(x) \approx f(x)$$

$$P_n(x) = f(x_0) + f'(x_0)(x-x_0) + \frac{f''(x_0)}{2!}(x-x_0)^2 + \frac{f'''(x_0)}{3!}(x-x_0)^3 + \dots$$

Ex 1: $f(x) = 2 - e^x + 2x$ $[0, 1]$

① $f'(x) = -e^x + 2$

② $-e^x + 2 = 0$

$\ln e^x = \ln 2$

$x = \ln 2 = 0.69$

$[1, 2]$

$f(1) = 2 - e^1 + 2(1) = 1.28$ ← absolute min

$f(2) = 2 - e^2 + 2(2) = 1.38$ ← absolute max

$f(0) = 2 - e^0 + 2(0) = 1$ ← absolute min

$f(1) = 2 - e^1 + 2(1) = 1.28$

$f(\ln 2) = 2 - e^{\ln 2} + 2(\ln 2) = 1.38$ ← absolute max

9. Let $f(x) = x^3$.

a. Find the second Taylor polynomial $P_2(x)$ about $x_0 = 0$.

① a $f(x) = x^3$ $x_0 = 0$

$f(0) = 0^3 = 0$

$f'(0) = 3(0)^2 = 0$

$f'(x) = 3x^2$

$f''(0) = 6(0) = 0$

$f''(x) = 6x$

$$P_2(x) = f(x_0) + f'(x_0)(x - x_0) + \frac{f''(x_0)(x - x_0)^2}{2!}$$

$= 0$

Note: You cannot change x^3 to a second degree

② $x_0 = 1$

$f(1) = 1^3 = 1$

$f'(1) = 3(1)^2 = 3$

$f''(1) = 6(1) = 6$

$$P_2(x) = f(x_0) + f'(x_0)(x - x_0) + \frac{f''(x_0)(x - x_0)^2}{2!}$$

$$= 1 + 3(x - 1) + \frac{6(x - 1)^2}{2!}$$

$$= 1 + 3x - 3 + 3(x^2 - 2x + 1)$$

$$= 1 + 3x - 3 + 3x^2 - 6x + 3$$

$$= 3x^2 - 3x + 1$$

7. Show that $f'(x)$ is 0 at least once in the given intervals. (Rolle's theorem)

b. $f(x) = (x-1)\tan x + x \sin \pi x$, $[0, 1]$

7b $f(x) = (x-1)\tan x + x \sin \pi x$ $[0, 1]$

Always right
this no matter
what the function
is about

Continuous in $[0, 1]$
derivative exists in $(0, 1)$

$$f(0) = (0-1)\tan 0 + 0 \sin(\pi \cdot 0) = 0$$

$$f(1) = (1-1)\tan 1 + 1 \sin(\pi \cdot 1) = 0$$

$$f(0) = f(1)$$

By Rolle's theorem there exists a
point $c \in [0, 1]$ such that $f'(c) = 0$

1. Show that the following equations have at least one solution in the given intervals.

a. $x \cos x - 2x^2 + 3x - 1 = 0$, $[0.2, 0.3]$ and $[1.2, 1.3]$

By IVT $f(0.2) = -0.28$
 $f(0.3) = 0.0066$

$f(a)$ and $f(b)$ are opposite in sign therefore
there exists a $c \in [0.2, 0.3]$ such that $f(c) = 0$
/// $[1.2, 1.3]$